

Table S1. Statistical Performance Characteristics of QSAR Models for HDAC8 Inhibitors
Developed Using the OCHEM Web Platform

Descriptors	Method	Training sample, 5-fold CV		Test sample	
				All compounds	
		Q^2_{cv}	RMSE	Q^2_{ts}	RMSE
ALogPS, OEstate	RFR	0.53	0.58	0.57	0.56
	ASNN	0.41	0.65	0.48	0.61
	KNN	0.42	0.65	0.43	0.64
	MLRA	0.24	0.74	0.28	0.73
	DNN	0.43	0.64	0.51	0.60
	XGBOOST	0.51	0.60	0.57	0.56
CDK	RFR	0.49	0.61	0.53	0.59
	ASNN	0.36	0.68	0.49	0.61
	KNN	0.43	0.64	0.47	0.62
	MLRA	0.25	0.74	0.26	0.73
	DNN	0.41	0.66	0.48	0.61
	XGBOOST	0.50	0.61	0.56	0.57
Dragon	RFR	0.49	0.61	0.5	0.61
	ASNN	0.34	0.69	0.46	0.62
	KNN	0.48	0.62	0.51	0.59
	MLRA	0	1.60	0.027	0.84
	DNN	0.33	0.70	0.37	0.68
	XGBOOST	0.46	0.62	0.51	0.60
Fragmentor (length: 2-4)	RFR	0.56	0.56	0.6	0.54
	ASNN	0.38	0.67	0.49	0.61
	KNN	0.41	0.66	0.43	0.64
	MLRA	0.21	0.75	0.23	0.75
	DNN	0.40	0.66	0.47	0.62
	XGBOOST	0.53	0.59	0.60	0.54
MOLD2	RFR	0.49	0.61	0.50	0.60
	ASNN	0.32	0.70	0.51	0.60
	KNN	0.43	0.64	0.49	0.61
	MLRA	0.25	0.74	0.31	0.71
	DNN	0.40	0.66	0.52	0.59
	XGBOOST	0.48	0.62	0.51	0.60
MORDRED	RFR	0.50	0.60	0.54	0.58
	ASNN	0.38	0.67	0.49	0.61
	KNN	0.47	0.62	0.50	0.60

QNPR	MLRA	0.12	0.80	0.28	0.72
	DNN	0.33	0.70	0.48	0.61
	XGBOOST	0.50	0.60	0.57	0.56
	RFR	0.53	0.58	0.56	0.57
	ASNN	0.31	0.71	0.32	0.70
	KNN	0.37	0.68	0.43	0.64
	MLRA	0.22	0.75	0.24	0.74
	DNN	0.30	0.71	0.38	0.67
	XGBOOST	0.48	0.61	0.54	0.58
	RFR	0.55	0.57	0.57	0.56
ECFP4	ASNN	0.41	0.65	0.50	0.60
	KNN	0.51	0.60	0.54	0.58
	MLRA	0.12	0.80	0.26	0.73
	DNN	0.32	0.70	0.40	0.66
	XGBOOST	0.46	0.62	0.50	0.60
	RFR	0.50	0.6	0.51	0.60
	ASNN	0.46	0.63	0.56	0.56
	KNN	0.49	0.61	0.52	0.59
	MLRA	0.14	0.79	0.28	0.72
	DNN	0.39	0.66	0.46	0.63
RDKit	XGBOOST	0.50	0.60	0.48	0.61
	RFR	0.50	0.60	0.50	0.60
	ASNN	0.36	0.68	0.50	0.60
	KNN	0.47	0.62	0.52	0.59
	MLRA	0	1.10	0	0.93
	DNN	0.31	0.71	0.32	0.71
	XGBOOST	0.49	0.61	0.49	0.61
	AttFP	0.41	0.66	0.51	0.59
	Chemprop	0.38	0.67	0.49	0.61
	TRANSNNI	0.57	0.56	0.62	0.52
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